SageMaker Lessons Learned + Deployment Cheat Sheet (Parameterized Edition)

Updated operational cheat sheet incorporating deployment failures, packaging fixes, environment isolation, and r

Core Parameter Block

```
REGION="us-east-2"
ROLE_ARN="arn:aws:iam::219217499832:role/SageMakerExecutionRole"
BUCKET="xt-benefits-model-bucket-fjg"
MODEL_BASE="sdoh-classifier"
VERSION="v1"
PREFIX="${MODEL_BASE}/${VERSION}"
MODEL_NAME="${MODEL_BASE}-${VERSION}"
ENDPOINT_CONFIG_NAME="${MODEL_BASE}-config"
ENDPOINT_NAME="${MODEL_BASE}-endpoint"
PKG_NAME="${MODEL_BASE}_sm_pkg"
MODEL_ARCHIVE="model.tar.gz"
MODEL_URI="s3://${BUCKET}/${PREFIX}/${MODEL_ARCHIVE}"
IMAGE_ACCOUNT="763104351884"
IMAGE_REPO="pytorch-inference"
IMAGE_TAG="2.0.1-cpu-py310-ubuntu20.04-sagemaker"
IMAGE="${IMAGE_ACCOUNT}.dkr.ecr.${REGION}.amazonaws.com/${IMAGE_REPO}:${IMAGE_TAG}"
VARIANT_NAME="AllTraffic"
INSTANCE_COUNT=1
INSTANCE_TYPE="ml.m5.large"
```

Packaging

```
rm -rf "/tmp/${PKG_NAME}"
mkdir -p "/tmp/${PKG_NAME}/code"
cp model/* "/tmp/${PKG_NAME}/"
cp code/*.py "/tmp/${PKG_NAME}/code/"
cp code/requirements.txt "/tmp/${PKG_NAME}/code/"
tar -czf "${MODEL_ARCHIVE}" -C "/tmp/${PKG_NAME}" .
tar -tzf "${MODEL_ARCHIVE}"
```

Create Model

```
aws s3 cp "${MODEL_ARCHIVE}" "${MODEL_URI}" --region "${REGION}"
aws sagemaker create-model --model-name "${MODEL_NAME}" --execution-role-arn "${ROLE_ARN}" --primary-container "
```

Create Endpoint

```
aws sagemaker create-endpoint-config --endpoint-config-name "${ENDPOINT_CONFIG_NAME}" --production-variants "Var
aws sagemaker create-endpoint --endpoint-name "${ENDPOINT_NAME}" --endpoint-config-name "${ENDPOINT_CONFIG_NAME}
```

Lessons Learned

Reuse one role, bucket, and AWS config. Separate venvs for GB and DL models. Match training and serving versions